

Creature Customizer Class Methods:

-all methods are get/set/remove

Creature Class Methods:

```
1. METHOD addDebuff(debuff: Debuff): void
    IF this.creature.resistances DOES NOT CONTAIN debuff THEN
        this.creature.debuffs.add(debuff)
    ELSE
        display blocked debuff message
    END IF
END METHOD

2. METHOD removeDebuff(debuff: Debuff): void
    this.creature.animations.happyAnimation.play()
    this.creature.debuffs.remove(debuff)
END METHOD

3. METHOD attack(): Integer
    selectedEnemy.health -= creature.damage
    this.creature.animations.attackAnimation.play()

    totalDamageDealt = this.creature.baseDamage

    selectedEnemy.armor -= totalDamageDealt

    healthDamageDealt = 0
    IF selectedEnemy.armor < 0 THEN
        healthDamageDealt = ABS(selectedEnemy.armor)
        selectedEnemy.armor = 0
    END IF

    IF healthDamageDealt > 0 THEN
        selectedEnemy.health -= healthDamageDealt
    END IF

    IF selectedEnemy.health <= 0 THEN
        selectedEnemy.removeFromTurnStack()
        selectedEnemy.animations.deathAnimation.play()
    END IF
```

```
    RETURN totalDamageDealt  
END METHOD
```

```
4. METHOD getLevelUpReward(): void  
    points = this.creature.level  
    boolean isDoneLeveling = false  
    WHILE isDoneLeveling == false  
        IF points > 0 THEN  
            pickTraitUpgrade()  
            points--  
        END IF  
        IF player wants to remove points from trait && trait points > 0 THEN  
            removeTraitUpgrade()  
            points++  
        END IF  
    END WHILE  
END METHOD
```

Player Class Methods:

```
1. METHOD addDebuff(debuff: Debuff): void  
    IF this.player.resistances DOES NOT CONTAIN debuff THEN  
        player.animations.sadAnimation.play()  
        this.player.debuffs.add(debuff)  
    ELSE  
        display blocked debuff message  
    END IF  
END METHOD
```

```
2. METHOD removeDebuff(debuff: Debuff): void  
    this.player.animations.happyAnimation.play()  
    this.player.debuffs.remove(debuff)  
END METHOD
```

```
3. METHOD useItem(item: Item, friendlyEntity: FriendlyEntity)  
    IF item.itemType == HEALING THEN  
        targetEntity.heal(item.healValue)  
        removeItem(item)  
    END IF  
END METHOD
```

```
4. METHOD heal(healValue: Integer): void
    this.player.health += healValue
    IF this.player.health > this.player.maxHealth
        This.player.health = this.player.maxHealth
    END IF
END METHOD
```

```
5. METHOD attack(): Integer
    player.animations.attackAnimation.play()

    totalDamageDealt = this.player.baseDamage

    selectedEnemy.armor -= totalDamageDealt

    healthDamageDealt = 0
    IF selectedEnemy.armor < 0 THEN
        healthDamageDealt = ABS(selectedEnemy.armor)
        selectedEnemy.armor = 0
    END IF

    IF healthDamageDealt > 0 THEN
        selectedEnemy.health -= healthDamageDealt
    END IF

    IF selectedEnemy.health <= 0 THEN
        selectedEnemy.removeFromTurnStack()
        selectedEnemy.animations.deathAnimation.play()
    END IF
    RETURN totalDamageDealt
END METHOD
```

EnemyEntity Class Methods:

```
1. METHOD addDebuff(debuff: Debuff): void
    this.enemyEntity.animations.angryAnimation.play()
    this.enemyEntity.debuffs.add(debuff)
END METHOD
```

```
2. METHOD removeDebuff(debuff: Debuff): void
    this.enemyEntity.animations.tauntAnimation.play()
    this.enemyEntity.debuffs.remove(debuff)
END METHOD
```

```
3. METHOD attack(): Integer
    target = findOptimalTarget()
    totalDamageDealt = this.enemyEntity.baseDamage

    target.armor -= totalDamageDealt

    healthDamageDealt = 0
    IF target.armor < 0 THEN
        healthDamageDealt = ABS(target.armor)
        target.armor = 0
    END IF

    IF healthDamageDealt > 0 THEN
        target.health -= healthDamageDealt
    END IF

    IF target.health <= 0 THEN
        target.removeFromTurnStack()
        target.animations.deathAnimation.play()
    END IF

    IF target.isPlayer() THEN
        endCombat()
    END IF

    RETURN totalDamageDealt

END METHOD
```

HatcheryHandler Class Methods:

```
1. METHOD feed(selectedFood: Food) Integer
    this.creature.hunger -= selectedFood.hungerRestoreValue
    removeItem(selectedFood)
    RETURN selectedFood.hungerRestoreValue
END METHOD
```

```
2. METHOD heal(item: Item): Integer
    this.creature.health += item.healValue
    IF this.creature.health > this.creature.maxHealth
        This.creature.health = this.creature.maxHealth
    END IF
    RETURN item.healValue
END METHOD
```

```
3. METHOD hatchEgg(selectedEgg: Egg): Creature
    Creature creature = selectedEgg.getRandomCreature(selectedEgg.eggType)
    addCreature(creature)
    RETURN creature
END METHOD
```

Rest of methods in hatchery handler are basic CRUD

Combat Handler Class Methods:

```
1. METHOD passTurn(): void
    turnStack[0].endTurn()
    currentTurnOwner = turnStack.pop()
    currentTurnOwner.startTurn()
END METHOD

2. METHOD giveDebuff(target: Entity, debuff: Debuff): void
    target.addDebuff(debuff)
END METHOD

3. METHOD getCombatReward(): CombatReward
    CombatReward reward = new CombatReward(level)
    IF NOT level.alreadyCompleted THEN
        RETURN reward;
    END IF

    RETURN null
END METHOD
```

-Rest of the methods are basic updates

Scene Manager Class Methods:

```

1. METHOD enterCombat(): void
    IF selectedCreatures.length > 0 AND verifyCreatureSelection(selectedCreatures) == true
    THEN
        FOR scene in openScenes
            scene.close()
        END FOR

        combatScene.init()
    END IF
END METHOD

```

```

2. METHOD verifyCreatureSelection(selectedCreatures: List<Creature>): Boolean
    FOR creature in selectedCreatures
        IF creature.health <= 0
            RETURN false
        END IF

        IF creature.isSick
            RETURN false
        END IF

        IF creature.isExhausted
            RETURN false
        END IF

    END FOR

    RETURN true
END METHOD

```

-rest of methods in scene manager are basic CRUD

Combat Reward class methods:

```

1. METHOD getRewards(): Map<Item, Integer>
    Map<Item, Integer> itemToCount = generateRewardsFromLevel(currentLevelNumber)
    RETURN itemToCount
END METHOD

```

- All other methods are basic CRUD

EquipmentManager class methods:

- all methods are basic CRUD

Egg Class Methods:

1. METHOD warm()

```
IF this.warmth >= 100 THEN
    return
END IF
```

```
ELSE IF this.warmth <= 0 THEN
    Creature in egg dies
    removeEgg()
END IF
```

```
ELSE
    creature.warmth += 10
END ELSE
```

END METHOD

2. METHOD getRandomCreature(eggType: EggType)

```
randomNum = RANDOM_INT(0, creatureEggsList[eggType].length)
return creatureEggs[type][randomNum]
```

END METHOD